# ETALE CHARTS OVER LOCALLY NOETHERIAN SPACES SEPARATE POINTS IN THE SAME FIBRE

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 1533-1541, immediately following the statement at 1525-1531.

*Editorial note.* The phrase 'By dimension theory for Noetherian local rings we conclude u = u' (source lines 1538-1541) compresses the dimension-gap argument; the expansion in proof_roadmap is the reviewer's checklist.

*Editorial note.* One cross-chapter section pointer (Schemes) and one cross-chapter lemma pointer (Properties of Spaces) are recorded in assembly.external_references.

*Allowed standard prerequisites (editorial).*

- Local rings along specializations are localizations. Schemes, Section schemes-section-points, cited at proof lines 1533-1535. u' ~ u in U gives O_{U,u'} is a localization of O_{U,u}.
- Local ring dimension is etale invariant up the specialization. Properties of Spaces, Lemma spaces-properties-lemma-pre-dimension-local-ring, cited at proof lines 1536-1537. dim(O_{U,u}) = dim(O_{U,u'}) – the equality that forces the specialization to be trivial.

*Source reference.* Original reference 01J5: focused-seeds/geometry/S9-decent-no-specializations-same-point-noetherian/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0BAM: focused-seeds/geometry/S9-decent-no-specializations-same-point-noetherian/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. Let $U \to X$ be an étale morphism from a scheme to $X$. Assume $u, u' \in |U|$ map to the same point $x$ of $|X|$, and $u' \rightsquigarrow u$. If $X$ is locally Noetherian, then $u = u'$.*

*Proof.* The discussion in Schemes, Section 01J5 shows that $\mathcal{O}_{U,u'}$ is a localization of the Noetherian local ring $\mathcal{O}_{U,u}$. By Properties of Spaces, Lemma 0BAM we have $\dim(\mathcal{O}_{U,u}) = \dim(\mathcal{O}_{U,u'})$. By dimension theory for Noetherian local rings we conclude $u = u'$. □

*Date*: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.